One-Dimensional Linear Advection

Finite-Volume Discretization, Explicit Time Integration, and Stability Analysis

Arian Shahnoori

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Abstract

A numerical solver is developed for the one-dimensional linear advection equation with constant transport velocity. The equation is discretized in space using a finite-volume formulation, with first-order upwind, second-order upwind, and second-order central-difference schemes. Time integration is performed using the explicit midpoint Runge–Kutta method (RK2) and the classical fourth-order Runge–Kutta method (RK4). Numerical solutions are verified against the analytical solution, and grid-refinement studies are used to assess spatial accuracy and the observed convergence order. Fourier-mode analysis is used to investigate the linear stability of selected space–time discretizations and to derive their limiting Courant–Friedrichs–Lewy (CFL) numbers. Further computational studies examine the effects of spatial discretization, CFL number, and temporal integration order, including the role of spatial-error dominance in the observed convergence behavior.

Problem	One-Dimensional Linear Advection
Governing Equation	First-Order Linear Advection Equation
Transport Velocity	Constant, $u = 2$
Spatial Discretization	Finite-Volume Method
Spatial Schemes	First-Order Upwind, Second-Order Upwind, Second-Order Central
Time Integration	Explicit Midpoint RK2, Classical RK4
Stability Analysis	Fourier-Mode (Von Neumann) Analysis
Verification	Analytical Solution and Grid Refinement
Programming Language	Python

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1. Introduction and Problem Definition

1.1 Governing Equation

The classical one-dimensional wave equation is a second-order hyperbolic partial differential equation (PDE):

$$\frac{\partial^2 T}{\partial t^2} = c^2 \frac{\partial^2 T}{\partial x^2}, \tag{1}$$

where T is the dependent variable, x is the spatial coordinate, t is time, and c is the wave propagation speed. The equation is second-order because its highest derivatives are of second order, and hyperbolic because it describes disturbances that propagate through the domain at finite speeds.

The present study focuses on the first-order linear advection equation,

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} = 0, \tag{2}$$

where u is a constant transport velocity. This equation describes the transport of a quantity T through a domain without changing its amplitude along a characteristic. It is also known as the linear convection equation. It is first-order because it contains only first-order derivatives and linear because the dependent variable and its derivatives appear linearly.

Although both equations describe wave propagation, they represent different mathematical models. The second-order wave equation describes phenomena such as acoustic wave propagation, whereas the first-order advection equation models the transport of a quantity at a prescribed velocity. In this project, the advection equation is the governing equation used to investigate the accuracy and stability of numerical discretization methods.

Since the advection equation contains both temporal and spatial derivatives, its numerical solution requires discretization in both time and space. Before solving the PDE, a time-dependent ordinary differential equation (ODE) is also considered to verify the time-integration method independently of spatial discretization.

1.2 Computational Domain and Analytical Solution

The computational domain is one-dimensional, with spatial extent

$$0 \leq x \leq L, \qquad L = 1. \tag{3}$$

The transport velocity is constant and positive, $u = 2$, indicating that the transported quantity moves from left to right. The governing equation is subject to the initial condition

$$T(x, 0) = 0, \qquad 0 \leq x \leq 1, \tag{4}$$

and the time-dependent inlet boundary condition

$$T(0, t) = \sin(2\pi t), \qquad t \geq 0. \tag{5}$$

The initial field is zero throughout the domain, while the inlet introduces a sinusoidal signal that propagates downstream with velocity u.

For these conditions, the analytical solution is

$$T(x,t) = \begin{cases} \sin\left[2\pi\left(t - \dfrac{x}{u}\right)\right], & x \le ut, \\ 0, & x > ut. \end{cases} \tag{6}$$

The condition $x = ut$ describes the advancing front of the transported signal. Ahead of this front, the quantity remains zero because the inlet disturbance has not yet reached that part of the domain. Behind the front, the solution corresponds to the sinusoidal inlet signal evaluated at the time when it entered the domain.

This analytical solution serves as a reference for assessing the accuracy of the numerical results. The study compares different spatial discretization schemes and time-integration methods, examines their stability as a function of the Courant number, and investigates how spatial and temporal discretization errors affect the overall numerical solution.

2. Numerical Implementation and Code Architecture

2.1 Modular Software Structure

The numerical solver was implemented in Python using a modular structure that separates the numerical algorithms from the experimental setup and visualization. The main objective is to make the implementation easier to understand, verify, and extend. In particular, different spatial discretization schemes and time-integration methods can be combined without duplicating the complete solver.

The source code is organized into modules according to their responsibilities. The `spatial_operators.py` module contains the spatially discretized right-hand sides for the second-order upwind and second-order central schemes. The `time_integrators.py` module implements the explicit midpoint Runge–Kutta method (RK2) and the classical fourth-order Runge–Kutta method (RK4). The `solver.py` module coordinates these components by defining the computational grid, determining the time-step size, initializing the solution, and advancing it to the final time.

Additional modules provide the analytical solution, error calculations, stability analysis, and visualization functions. Individual numerical experiments are implemented as separate scripts in the `studies` directory. Each study specifies its parameters, calls the solver, and analyzes and plots the results. Finally, `main.py` serves as the entry point for executing the selected studies.

This design separates the implementation of numerical methods from the setup of individual experiments. For example, changing the time integrator from RK2 to RK4 does not require rewriting the spatial discretization or the time-marching loop.

2.2 Separation of Spatial and Temporal Operators

Following spatial discretization, the governing equation is converted into a system of ordinary differential equations (ODEs),

$$\frac{d\mathbf{T}}{dt} = \mathbf{R}(\mathbf{T}, t), \tag{7}$$

where $\mathbf{T}$ is the vector of cell-centred solution values and $\mathbf{R}$ denotes the spatially discretized right-hand side. The temporal integration method advances this system in time, while the spatial operator determines the right-hand-side value for a given solution state.

In the implementation, the spatial operator is selected using the `spatial_scheme` parameter. A Python dictionary associates each scheme name with its corresponding function:

```
spatial_operators = {
    "2nd_upwind": rhs_second_order_upwind,
    "2nd_central": rhs_second_order_central,
}

spatial_operator = spatial_operators[spatial_scheme]
```

For example, if `spatial_scheme` is `"2nd_upwind"`, the variable `spatial_operator` refers to `rhs_second_order_upwind`. If `"2nd_central"` is selected, it instead refers to `rhs_second_order_central`. The dictionary stores references to functions, not the numerical values returned by those functions. Assigning a function to another variable therefore provides another name through which the same function can be called; it does not execute or modify the function.

The time integrator is selected in the same way:

```
time_integrators = {
    "rk2": rk2_midpoint_step,
    "rk4": rk4_classical_step,
}

time_stepper = time_integrators[temporal_scheme]
```

Here, `time_stepper` refers to the function associated with the selected temporal scheme. Both integrators use the same interface: the current solution, current time, time-step size, and a function that evaluates the right-hand side. This common interface lets the solver switch between RK2 and RK4 without changing the time-marching loop.

2.3 Passing the Right-Hand-Side Function to the Time Integrator

A central feature of the implementation is that a function is passed as an argument to another function. In Python, functions are objects that can be assigned to variables, passed to other functions, and called later. Such an object is commonly described as a *callable*.

The selected spatial operator requires the solution state, time, grid spacing, and transport velocity. To expose it through the common interface expected by the time integrators, the solver defines a nested function:

```
def rhs(state, time):
    return spatial_operator(state, time, dx, u)
```

The variables `state` and `time` are parameters of `rhs`; their values are supplied each time the function is called. The variables `dx`, `u`, and `spatial_operator` are defined in the surrounding `solve_advection` function and are accessible within the nested function. This allows `rhs` to present a consistent two-argument interface while forwarding the necessary grid and physical parameters to the selected spatial operator.

The function is then passed to the chosen time integrator:

```
T = time_stepper(T, t_n, dt, rhs)
```

At this stage, `time_stepper` refers to either `rk2_midpoint_step` or `rk4_classical_step`. Since both integrators accept the same argument structure, the call remains valid when the temporal scheme is changed.

For the explicit midpoint RK2 method, the supplied right-hand-side function is evaluated twice:

```
k1 = rhs(T, t_n)

T_half = T + 0.5 * dt * k1

k2 = rhs(T_half, t_n + 0.5 * dt)
```

In the first evaluation, `state` receives the current solution $\mathbf{T}^n$ and `time` receives t_n. In the second evaluation, they receive the intermediate solution

$$\mathbf{T}^{n+1/2} = \mathbf{T}^n + \frac{\Delta t}{2}\,\mathbf{R}(\mathbf{T}^n, t_n) \tag{8}$$

and the midpoint time $t_n + \Delta t/2$, respectively. Each call to `rhs` forwards the current state and time to the selected spatial operator, together with the fixed values of Δx and u. The same spatial discretization can therefore be evaluated at intermediate states without embedding its implementation inside the time integrator.

The classical RK4 method follows the same principle, but evaluates the right-hand side four times at its prescribed intermediate states and times. Thus, the time integrator controls when and how often the right-hand side is evaluated, while the spatial operator controls how each right-hand-side value is computed.

This function-passing approach keeps temporal integration independent of the details of spatial discretization. It also allows the available spatial and temporal schemes to be combined through the solver's input parameters.

2.4 Time Marching and Numerical Experiments

The computational domain is divided into uniformly spaced control volumes, and the solution is initialized using the zero initial condition, $T(x, 0) = 0$. The time-step size can be determined either from a prescribed Courant number or from an explicitly specified number of time steps.

When the Courant number is prescribed, the solver calculates a time-step limit from the requested value and selects an integer number of steps sufficient to reach the final time. When the number of steps is specified directly, the time-step size is calculated so that the simulation reaches the requested final time exactly.

The solution is then advanced iteratively. At each time step, the selected time integrator receives the current solution, current time, time-step size, and right-hand-side function. The updated solution becomes the starting state for the next step, and this process continues until the final time is reached.

The numerical experiments are defined separately from the solver. Each study specifies the discretization methods and simulation parameters relevant to its objective, such as grid resolution, Courant number, final time, or number of time steps. The computed results are then compared with the analytical solution, assessed using the selected error measures, and visualized using shared plotting functions.

This organization supports systematic comparisons of numerical accuracy and stability while keeping the underlying solver reusable across different experiments.

3. Stability Analysis

Numerical stability is a fundamental property of a time-marching scheme. Even when the governing equation has a bounded solution, discretization errors can grow with each time step and eventually dominate the numerical solution. For the linear advection equation, stability depends on the combination of spatial discretization and time integration, as well as on the selected time-step size.

The Courant number is defined as

$$C = \frac{u\Delta t}{\Delta x}, \tag{9}$$

where u is the transport velocity, Δt is the time-step size, and Δx is the cell width. This dimensionless number relates temporal and spatial resolution and provides a convenient parameter for analysing stability.

To investigate the stability of the numerical schemes, Fourier-mode analysis is applied to their spatially discretized equations. For a uniform grid and a constant-coefficient interior spatial operator, the solution is represented by Fourier modes of the form

$$T_i(t) = \widehat{T}(t)e^{\mathrm{i}i\phi}, \tag{10}$$

where ϕ is the dimensionless wave number and $\mathrm{i} = \sqrt{-1}$. Substitution into the semi-discrete equation transforms the spatial operator into a scalar eigenvalue problem. The time-integration method then determines how the amplitude of each mode changes between consecutive time levels.

Defining $z = \lambda\Delta t$, where λ is the spatial eigenvalue, the numerical evolution of a mode can be written as

$$\widehat{T}^{n+1} = G(z)\widehat{T}^{n}, \tag{11}$$

where $G(z)$ is the numerical amplification factor. The absolute-stability condition is

$$\boxed{|G(z)| \leq 1}. \tag{12}$$

If the amplification factor exceeds unity for a mode, that mode grows numerically. The following subsections examine this condition for second-order upwind discretization combined with RK2 and RK4, and for second-order central discretization combined with RK2.

3.1 RK2 with Second-Order Upwind Discretization

For the second-order upwind scheme, the semi-discrete equation for an interior cell is

$$\frac{dT_i}{dt} = -\frac{u}{2\Delta x}\left(3T_i - 4T_{i-1} + T_{i-2}\right). \tag{13}$$

Substituting a Fourier mode into this equation gives the dimensionless eigenvalue

$$\boxed{z = -\frac{C}{2}\left(3 - 4e^{-\mathrm{i}\phi} + e^{-2\mathrm{i}\phi}\right)}. \tag{14}$$

For the explicit midpoint RK2 method, the amplification factor is

$$G_{\mathrm{RK2}}(z) = 1 + z + \frac{z^2}{2}. \tag{15}$$

The stability region is determined by evaluating $|G_{\mathrm{RK2}}|$ over a range of Fourier phases and Courant numbers. The boundary $|G_{\mathrm{RK2}}| = 1$ separates stable and unstable modes.

The limiting Courant number can be identified by examining the highest-frequency mode, $\phi = \pi$. For this mode,

$$z = -4C, \tag{16}$$

and the amplification factor becomes

$$G_{\mathrm{RK2}} = 1 - 4C + 8C^2. \tag{17}$$

Applying the absolute-stability condition gives

$$|1 - 4C + 8C^2| \leq 1, \tag{18}$$

which, for $C \geq 0$, yields

$$\boxed{0 \leq C \leq 0.5}. \tag{19}$$

The full Fourier-mode analysis confirms that $C_{\max} = 0.5$ is the limiting Courant number for this spatial discretization combined with RK2. At this limit, the highest-frequency mode is marginally stable, while a larger Courant number produces at least one unstable mode. The resulting stability diagram provides a theoretical reference for selecting the time step. Its predictions are compared with numerical simulations in Section 4.

3.2 RK2 with Second-Order Central Discretization

The second-order central spatial discretization gives the semi-discrete equation

$$\frac{dT_i}{dt} = -\frac{u}{2\Delta x}\left(T_{i+1} - T_{i-1}\right). \tag{20}$$

Substitution of a Fourier mode gives the spatial eigenvalue

$$\lambda(\phi) = -\mathrm{i}\frac{u}{\Delta x}\sin\phi. \tag{21}$$

Consequently, the dimensionless eigenvalue is purely imaginary:

$$z = \lambda\Delta t = -\mathrm{i}C\sin\phi = \mathrm{i}y, \qquad y = -C\sin\phi. \tag{22}$$

Using the RK2 amplification factor in Equation (15) and substituting $z = \mathrm{i}y$ gives

$$G_{\mathrm{RK2}} = 1 - \frac{y^2}{2} + \mathrm{i}y. \tag{23}$$

Its magnitude satisfies

$$|G_{\mathrm{RK2}}|^2 = \left(1 - \frac{y^2}{2}\right)^2 + y^2 \tag{24}$$

$$= 1 + \frac{y^4}{4}. \tag{25}$$

Therefore,

$$\boxed{|G_{\mathrm{RK2}}|^2 = 1 + \frac{C^4\sin^4\phi}{4}}. \tag{26}$$

For any nonzero mode with $\sin\phi \neq 0$, the amplification factor exceeds unity for every positive Courant number. Thus, the combination of second-order central differencing and explicit midpoint RK2 does not satisfy the absolute-stability condition for all Fourier modes at any positive time-step size.

This result differs from the upwind scheme, whose spatial eigenvalues include a dissipative component that can be balanced against the time integrator's stability properties. For the central scheme, reducing the Courant number decreases the amplification of individual modes but does not eliminate the underlying instability.

The numerical consequences are examined in the results section. In particular, computed solutions can remain close to the analytical solution over a finite simulation interval even when the method is theoretically unstable. Such finite-time agreement should not be interpreted as proof of stability.

3.3 Classical RK4 with Second-Order Upwind Discretization

The classical fourth-order Runge–Kutta method (RK4) is analysed using the same second-order upwind spatial discretization. Consequently, the dimensionless spatial eigenvalue remains

$$z = -\frac{C}{2}\left(3 - 4e^{-\mathrm{i}\phi} + e^{-2\mathrm{i}\phi}\right). \tag{27}$$

The difference lies in the temporal amplification factor. For classical RK4,

$$G_{\mathrm{RK4}}(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}. \tag{28}$$

The stability condition is again

$$\boxed{|G_{\mathrm{RK4}}(z)| \leq 1}. \tag{29}$$

Evaluating this condition over a range of Fourier phases and Courant numbers produces the RK4 stability diagram. The limiting Courant number is approximately

$$\boxed{C_{\max} \approx 0.6963}. \tag{30}$$

This limit can be established by considering the highest-frequency mode, $\phi = \pi$, for which $z = -4C$. Along the negative real axis, the classical RK4 stability polynomial remains within its absolute-stability region down to approximately $z = -2.7853$. Thus,

$$C_{\max} \approx \frac{2.7853}{4} \approx 0.6963. \tag{31}$$

The full Fourier-mode analysis confirms that this mode determines the limiting Courant number for the second-order upwind interior stencil combined with classical RK4.

Compared with RK2, classical RK4 permits a larger stable Courant number for this spatial discretization. Nevertheless, the stability limit does not mean that the largest allowable time step necessarily provides the most accurate solution. Accuracy and stability are distinct considerations, and a suitable time step must account for both.

Overall, the stability analysis demonstrates that the time-integration method cannot be assessed independently of the spatial discretization. The second-order upwind scheme combined with RK2 is stable for $C \leq 0.5$, while its combination with classical RK4 permits a limiting Courant number of approximately 0.6963. In contrast, central differencing combined with explicit midpoint RK2 fails the absolute-stability condition for nonzero Fourier modes at every positive Courant number. These theoretical findings provide the basis for interpreting the numerical behaviour examined in the next section.

Applicability of the Fourier Stability Analysis. The Fourier stability analysis presented in this section assumes periodic boundary conditions, whereas the advection problem considered in this study has a prescribed time-dependent inlet condition. Therefore, the derived CFL limits characterize the stability of the interior spatial discretization combined with the selected time integrator, rather than providing a complete stability proof for the full numerical problem. The boundary treatment may also influence the stability of the complete discretized system. To assess the applicability of these predictions, the theoretical results were compared with numerical tests

using the actual solver and its prescribed inlet condition. For the RK2 scheme combined with second-order upwind discretization, the solution remained bounded at CFL numbers of (0.4) and (0.5), whereas instability was observed at approximately (0.5714), consistent with the predicted interior stability limit of ($C_{\max} = 0.5$). This agreement supports the use of the Fourier analysis as a practical stability criterion for the present problem, although it does not establish a general stability guarantee for the non-periodic system.

4. Results and Discussion

4.1 Grid-Convergence Verification of the Second-Order Upwind Scheme

The spatial accuracy of the second-order upwind scheme combined with explicit midpoint RK2 was assessed using a grid-refinement study. The simulations were performed up to $t = 1$, with a fixed Courant number of $C = 0.25$. Four spatial resolutions, consisting of 20, 40, 80, and 160 control volumes, were considered. At each resolution, the numerical solution was compared with the analytical solution, and the discrete RMS L^2 error was calculated.

The numerical results are summarized in Table 1.

Table 1: Grid-convergence study for second-order upwind discretization with RK2 at $t = 1$ and $C = 0.25$.

Control volumes	Δt	Time steps	CFL	RMS L^2 error
20	6.2500×10^{-3}	160	0.25	1.025270×10^{-2}
40	3.1250×10^{-3}	320	0.25	2.781684×10^{-3}
80	1.5625×10^{-3}	640	0.25	7.200084×10^{-4}
160	7.8125×10^{-4}	1280	0.25	1.828248×10^{-4}

The error decreases consistently as the grid is refined. Doubling the number of control volumes approximately reduces the error by a factor of four, as expected for second-order convergence. The log–log regression gives an observed convergence order of 1.9378, which is close to the theoretically expected order of two. The convergence behaviour is illustrated in Figure 1.

Since the Courant number is held fixed, both spatial and temporal resolutions change simultaneously. Therefore, this experiment assesses convergence of the combined numerical method rather than isolating spatial error alone. Nevertheless, the observed rate is consistent with the expected second-order accuracy.

4.2 Numerical Verification of the RK2 Stability Limit

The theoretical stability limit obtained from Fourier analysis was compared with numerical simulations using 40 control volumes and a final time of $t = 0.25$. Three time-step counts were considered: $N = 50$, $N = 40$, and $N = 35$. The corresponding time-step sizes and Courant numbers are shown in Table 2.

The Fourier stability map in Figure 2 identifies $C_{\max} = 0.5$ as the limiting Courant number for second-order upwind discretization combined with RK2. The corresponding numerical solutions are compared with the analytical solution in Figure 3.

The $N = 50$ case, corresponding to $C = 0.4$, remains within the predicted stable range. The

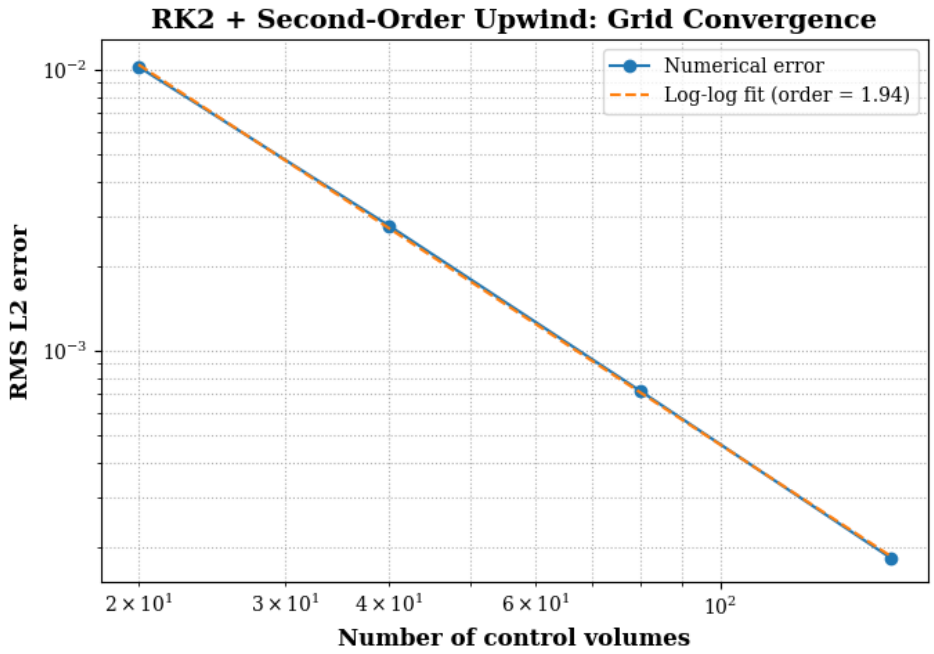


Figure 1: Grid convergence of the RMS L^2 error for RK2 combined with second-order upwind discretization at a fixed CFL number of 0.25.

Table 2: Time-step sizes and actual Courant numbers for the RK2 stability-verification study at $t = 0.25$.

Time steps	Δt	Actual CFL
50	5.000000×10^{-3}	0.4000
40	6.250000×10^{-3}	0.5000
35	7.142857×10^{-3}	0.5714

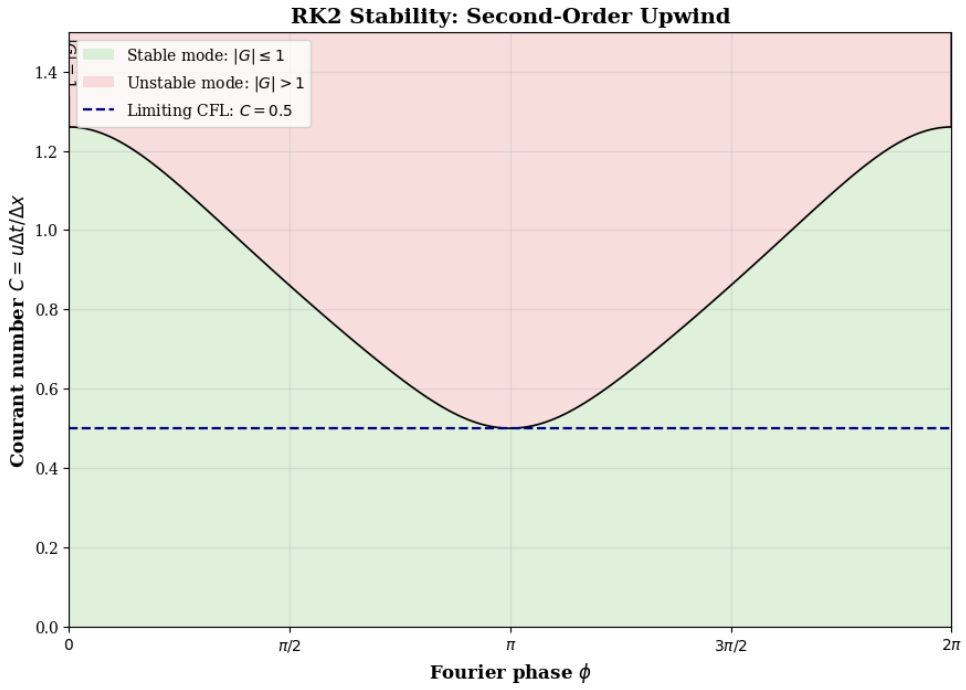


Figure 2: Fourier stability map for RK2 combined with second-order upwind discretization. The stability boundary is defined by $|G| = 1$, and the limiting Courant number is $C_{\max} = 0.5$.

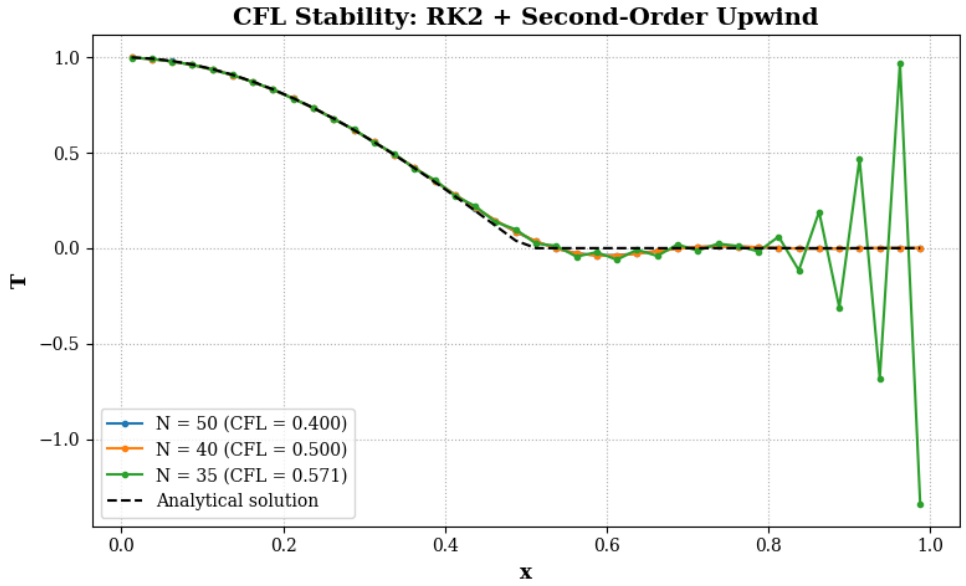


Figure 3: Comparison of the numerical and analytical solutions for the RK2 stability-verification cases at $t = 0.25$, using 40 control volumes and actual CFL numbers of 0.4, 0.5, and approximately 0.5714.

$N = 40$ case lies exactly at the theoretical stability boundary and remains bounded over the tested simulation interval. In contrast, the $N = 35$ case exceeds the stability limit, and the numerical solution exhibits unstable behaviour.

These observations are consistent with the Fourier stability analysis. The result at $C = 0.5$ represents a limiting case rather than a robust operating condition. In practical simulations, a Courant number below the theoretical limit provides a safety margin against numerical instability.

4.3 Grid Refinement with Second-Order Central Discretization

The grid-refinement procedure was also applied to the second-order central scheme combined with RK2, using a final time of $t = 1$, a Courant number of $C = 0.25$, and 20 and 40 control volumes. The results are given in Table 3.

Table 3: Grid-refinement study for second-order central discretization with RK2 at $t = 1$ and $C = 0.25$.

Control volumes	Δt	Time steps	CFL	RMS L^2 error
20	6.250000×10^{-3}	160	0.25	8.540923×10^{-3}
40	3.125000×10^{-3}	320	0.25	2.868139×10^{-3}

The errors decrease as the grid is refined. The log–log fit gives an observed convergence order of 1.5743, as illustrated in Figure 4.

Although this result indicates an improvement in accuracy with grid refinement, the measured order is below two. Furthermore, the estimate is based on only two resolutions, so it should not be considered a conclusive verification of second-order convergence. The theoretical stability analysis also shows that central differencing combined with explicit midpoint RK2 fails the absolute-stability condition for nonzero purely imaginary Fourier modes. Consequently, the

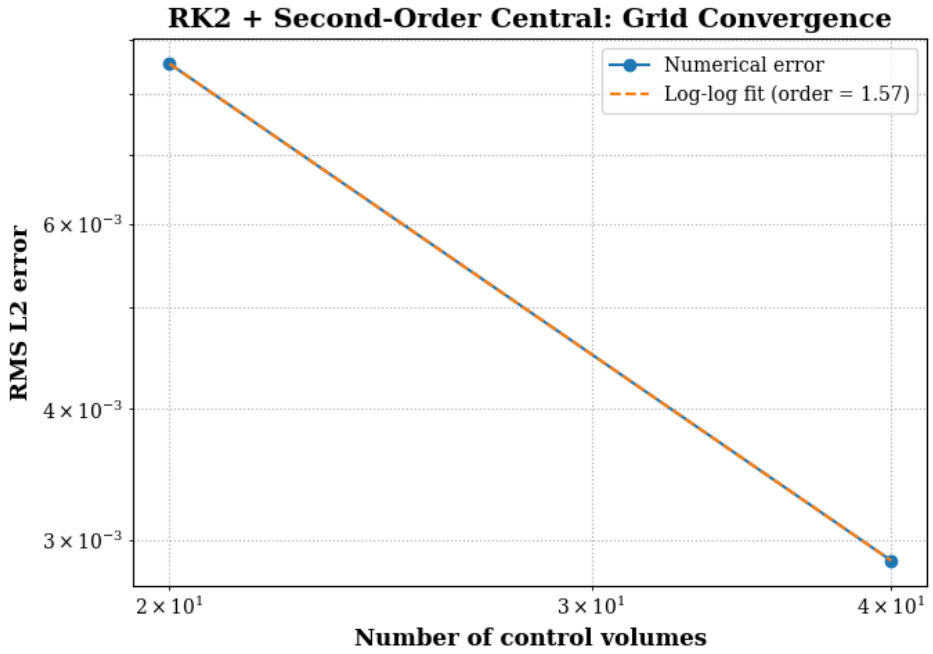


Figure 4: Grid-refinement results for RK2 combined with second-order central discretization at a fixed CFL number of 0.25.

observed error reduction must be interpreted as finite-resolution behaviour rather than proof of stable convergence.

4.4 Comparison of Central and Upwind Schemes at Different CFL Values

The second-order central and second-order upwind schemes were compared using 40 control volumes at $t = 1$, with requested Courant numbers of 0.7, 0.4, and 0.1. The actual CFL values were approximately 0.7018, 0.4000, and 0.1000, respectively. These cases were selected to examine the predicted stability properties of the two schemes under different time-step restrictions.

For the second-order upwind scheme, the stability analysis predicts a limiting Courant number of 0.5. Accordingly, the $C \approx 0.7$ case lies outside its stable range, whereas the $C = 0.4$ and $C = 0.1$ cases are within it. For central differencing combined with RK2, the Fourier analysis predicts amplification of nonzero modes for every positive Courant number, although the growth may be less apparent over a finite simulation.

The numerical and analytical solution comparisons are shown in Figures 5–7.

Diagnostic consistency check. The numerical diagnostics from the current implementation require verification before a definitive comparison can be reported. At all three CFL values, the logged RMS errors, maximum errors, and maximum solution magnitudes are identical for the two schemes. For example, at $C = 0.4$, both are reported to have an RMS error of 2.912544×10^{-3} and a maximum absolute solution value of 0.996774. Identical results are also reported for the other two cases, including the $C \approx 0.7$ case.

These results do not establish that the two schemes behave identically. Their spatial operators and theoretical stability properties are different. The scheme selection and right-hand-side

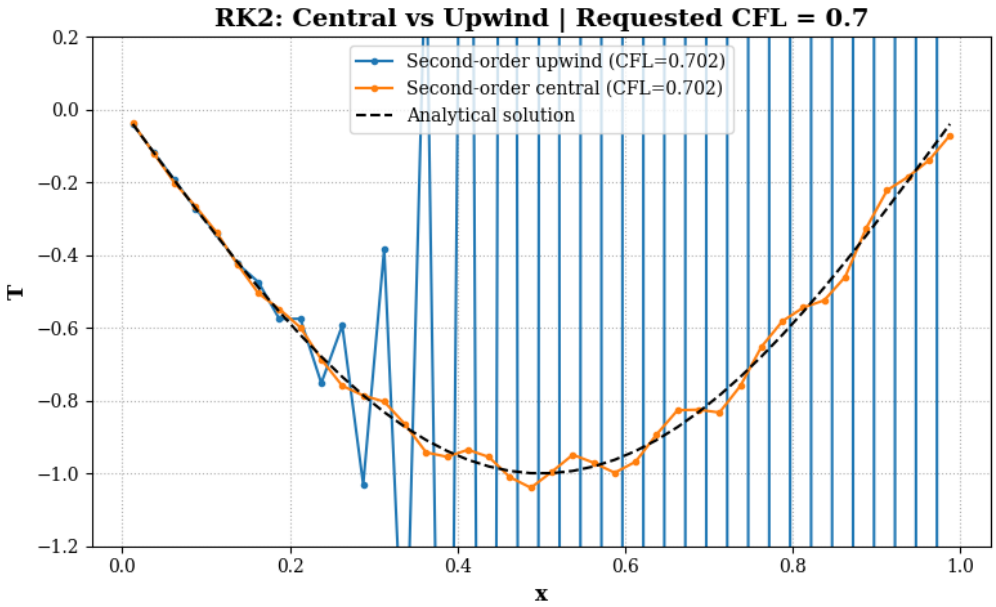


Figure 5: Intended comparison of second-order upwind and second-order central discretizations with RK2 at an actual CFL number of approximately 0.7018.

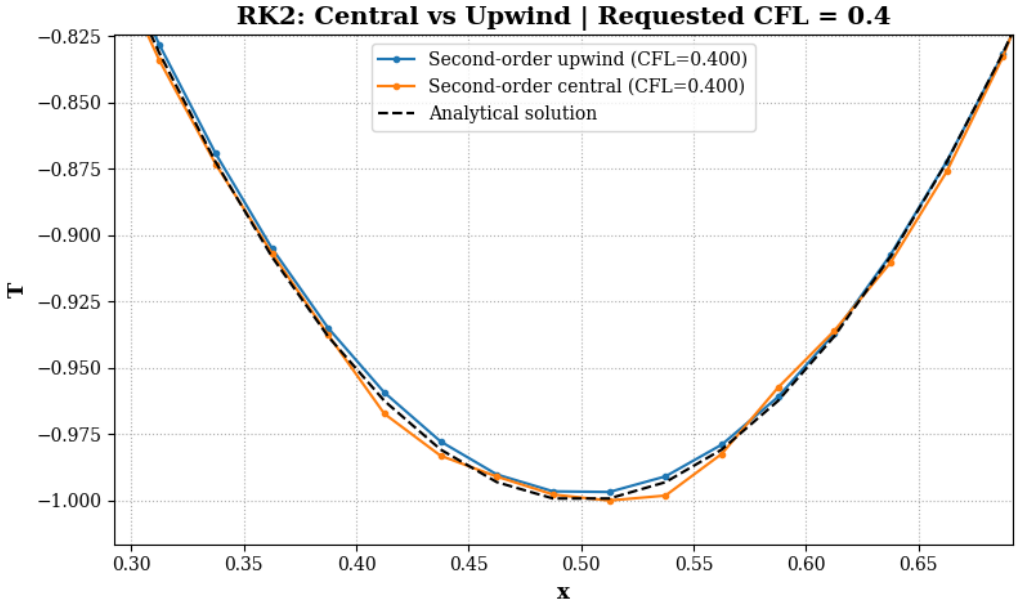


Figure 6: Intended comparison of second-order upwind and second-order central discretizations with RK2 at a CFL number of 0.4.

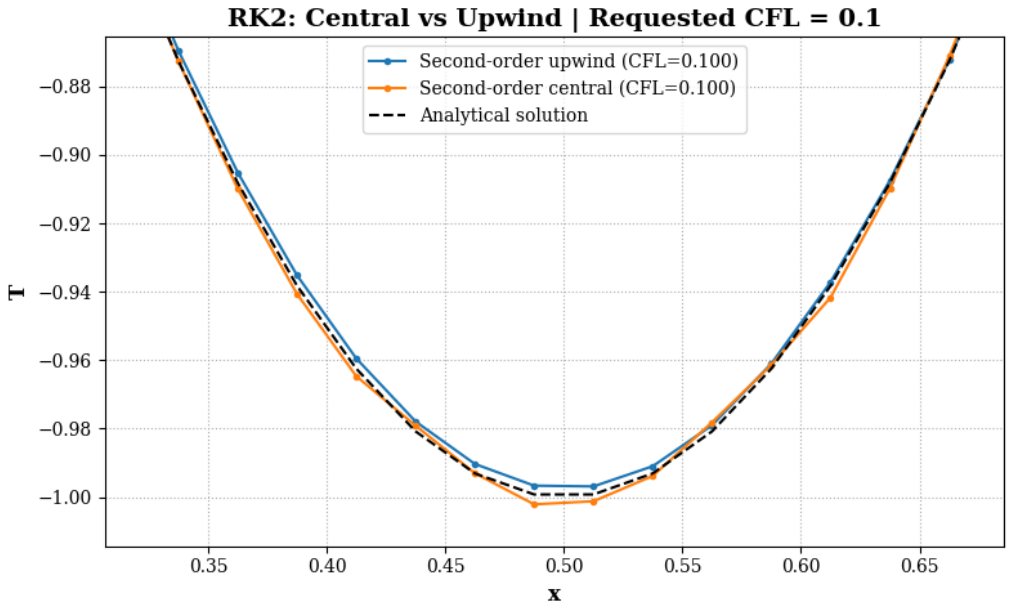


Figure 7: Intended comparison of second-order upwind and second-order central discretizations with RK2 at a CFL number of 0.1.

implementation should therefore be checked, and the simulations and figures regenerated before drawing conclusions from this comparison. Until this issue is resolved, Figures 5–7 should be treated as provisional outputs rather than evidence supporting a quantitative comparison.

4.5 Stability of Classical RK4 with Second-Order Upwind Discretization

The stability of classical RK4 combined with second-order upwind spatial discretization was examined by evaluating its amplification factor over a range of Fourier phases and Courant numbers. The resulting stability map is shown in Figure 8.

The theoretical limiting Courant number is approximately $C_{\max} = 0.6963$. In the numerical calculation, 601 Courant-number points were sampled over the interval $0 \leq C \leq 2$. The largest sampled Courant number satisfying the all-mode stability condition was 0.6933.

The small difference between these values is attributable to the finite resolution of the sampled Courant-number grid: the numerical search can identify only the largest stable value among the discrete samples. Increasing the sampling resolution would provide a closer numerical approximation to the theoretical limit.

The analysis shows that classical RK4 permits a higher limiting Courant number than RK2 for the same second-order upwind spatial discretization. This theoretical result guides the choice of time step for the accuracy comparison. It does not, however, imply that the largest stable Courant number necessarily yields the most accurate solution.

4.6 Effect of Temporal Integration Order on Overall Accuracy

The accuracy of midpoint RK2 and classical RK4 was compared using the same second-order upwind spatial discretization. Two experiments were performed to distinguish temporal error from the overall error associated with the spatial and temporal discretizations.

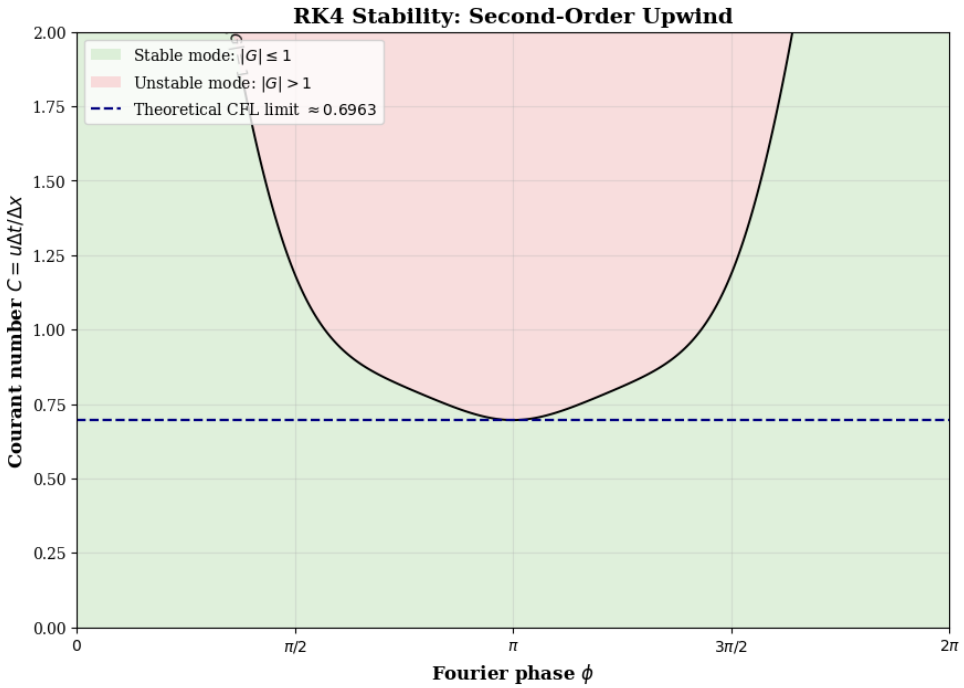


Figure 8: Fourier stability map for classical RK4 combined with second-order upwind discretization.

4.6.1 Temporal Refinement on a Fixed Grid

The first experiment used a fixed grid of 40 control volumes and progressively increased the number of time steps from 200 to 3200. The results are presented in Table 4.

Table 4: Temporal refinement on a fixed grid of 40 control volumes using second-order upwind spatial discretization.

Time steps	Δt	CFL	RK2 RMS L^2 error	RK4 RMS L^2 error
200	5.000000×10^{-3}	0.400	2.912544×10^{-3}	2.698545×10^{-3}
400	2.500000×10^{-3}	0.200	2.751668×10^{-3}	2.698561×10^{-3}
800	1.250000×10^{-3}	0.100	2.711795×10^{-3}	2.698562×10^{-3}
1600	6.250000×10^{-4}	0.050	2.701865×10^{-3}	2.698562×10^{-3}
3200	3.125000×10^{-4}	0.025	2.699387×10^{-3}	2.698562×10^{-3}

The RK2 error decreases with temporal refinement, approaching the nearly constant error obtained using RK4. The RK4 error remains approximately 2.6986×10^{-3} across the tested time steps, indicating that its temporal integration error is negligible relative to the error associated with the fixed spatial grid.

As the RK2 time-step size decreases, its temporal error becomes progressively less significant, and the total error approaches the same level. Thus, further reducing the time-step size yields diminishing improvements because the spatial discretization error becomes dominant.

4.6.2 Grid Refinement at a Fixed Courant Number

The second experiment examined grid refinement using a constant Courant number of $C = 0.05$. This small value reduces the temporal contribution to the error, allowing the overall convergence behaviour to be examined. Both time integrators were tested using the same spatial resolutions and time steps. The results are given in Table 5.

Table 5: Grid refinement at a fixed Courant number of $C = 0.05$ for second-order upwind discretization with RK2 and RK4.

Control volumes	Time steps	Δt	CFL	RK2 RMS L^2 error	RK4 RMS L^2 error
20	800	1.250000×10^{-3}	0.05	9.948121×10^{-3}	9.935563×10^{-3}
40	1600	6.250000×10^{-4}	0.05	2.701865×10^{-3}	2.698562×10^{-3}
80	3200	3.125000×10^{-4}	0.05	6.997847×10^{-4}	6.989454×10^{-4}
160	6400	1.562500×10^{-4}	0.05	1.777493×10^{-4}	1.775383×10^{-4}

The observed convergence orders are 1.9368 for both RK2 and RK4, consistent with second-order convergence of the combined numerical solution. The convergence curves are shown in Figure 9.

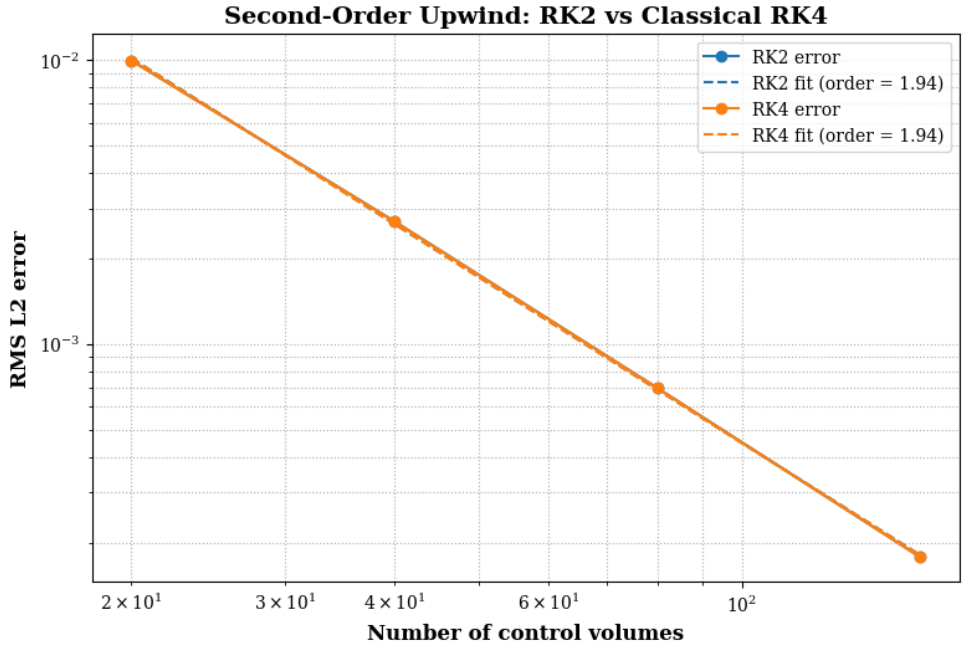


Figure 9: Grid-convergence comparison of RK2 and classical RK4 with second-order upwind spatial discretization at a fixed CFL number of 0.05.

The results demonstrate that increasing the temporal order does not necessarily increase the overall convergence order of a numerical solution. Although classical RK4 has fourth-order temporal accuracy, both methods exhibit approximately second-order overall convergence because they use the same second-order spatial discretization. When the temporal contribution to the error is small, the spatial error dominates the total error and determines the observed convergence rate.

Taken together, the two experiments distinguish the formal accuracy of a time integrator from that of the complete numerical method. Temporal refinement at a fixed grid reveals the error

floor associated with spatial discretization, while grid refinement at a sufficiently small Courant number confirms that both methods approach the expected second-order overall convergence rate.

5. Conclusion

This study developed and evaluated a modular finite-volume solver for the one-dimensional linear advection equation with a prescribed time-dependent inlet condition. Second-order upwind and second-order central spatial discretizations were considered in combination with the explicit midpoint Runge–Kutta method (RK2) and the classical fourth-order Runge–Kutta method (RK4). The analytical solution was used as a reference to assess numerical accuracy, while grid refinement, time-step refinement, and stability analysis were used to investigate the behavior of the numerical schemes.

For the second-order upwind scheme combined with RK2, grid refinement at a constant CFL number of 0.25 produced an observed convergence order of approximately 1.94, consistent with the expected second-order accuracy. Time-step refinement on a fixed grid further showed that the RK2 error approached the spatial discretization error as the time step decreased. With RK4, the temporal error was already sufficiently small for the spatial error to dominate. Consequently, grid refinement at a constant CFL number yielded an observed order of approximately 1.94 for both time integrators. These results demonstrate that increasing the temporal order does not necessarily increase the overall accuracy when the spatial discretization is the dominant source of error.

The Fourier stability analysis of the second-order upwind discretization gave a theoretical CFL limit of $C \leq 0.5$ for RK2. Tests using the full solver were consistent with this prediction: the solution remained bounded at CFL numbers of 0.4 and 0.5, whereas instability was observed at approximately 0.571. The result at the limiting value should nevertheless be interpreted as marginal stability rather than as a robust operating condition. For RK4, the corresponding theoretical limit was approximately $C_{\max} = 0.6963$, allowing a larger time step than RK2 for the same spatial grid. A numerical search yielded a slightly lower estimate because of its finite resolution.

The stability analysis also showed that the second-order central discretization combined with RK2 is unstable for nonzero Fourier modes at any positive CFL number. Therefore, a small time step or apparently acceptable behavior over a short simulation does not establish the long-time stability of this combination.

Overall, the second-order upwind scheme provides a suitable baseline for the present advection problem. RK2 offers a straightforward and effective time integrator when used within its stability limit, whereas RK4 provides a wider stability range and smaller temporal error. In either case, the accuracy of the complete solver remains constrained by the second-order spatial discretization. Finally, the Fourier analysis characterizes the stability of the interior discretization under periodic-mode assumptions; confirming stability for the non-periodic inlet-boundary problem requires consideration of the actual boundary treatment as well.

A. Fourier Stability Analysis of RK2 with Second-Order Upwind

A.1 Derivation of the Spatial Eigenvalue

The second-order upwind spatial discretization of the one-dimensional linear advection equation gives the semi-discrete equation

$$\frac{dT_j}{dt} = -\frac{u}{2\Delta x}\left(3T_j - 4T_{j-1} + T_{j-2}\right), \tag{32}$$

where j denotes the cell index. To analyse the stability of the interior spatial operator, consider a Fourier mode of the form

$$T_j(t) = \hat{T}(t)e^{\mathrm{i}j\phi}, \tag{33}$$

where $\hat{T}(t)$ is the mode amplitude, ϕ is the dimensionless wave number, and $\mathrm{i} = \sqrt{-1}$. The neighbouring values are

$$T_{j-1} = T_j e^{-\mathrm{i}\phi}, \qquad T_{j-2} = T_j e^{-2\mathrm{i}\phi}. \tag{34}$$

Substituting these expressions into the semi-discrete equation yields

$$\frac{d\hat{T}}{dt} = \lambda(\phi)\hat{T}, \tag{35}$$

where the spatial eigenvalue is

$$\boxed{\lambda(\phi) = -\frac{u}{2\Delta x}\left(3 - 4e^{-\mathrm{i}\phi} + e^{-2\mathrm{i}\phi}\right)}. \tag{36}$$

Introducing the Courant number

$$C = \frac{u\Delta t}{\Delta x}, \tag{37}$$

and defining the dimensionless eigenvalue $z = \lambda\Delta t = x + \mathrm{i}y$, we obtain

$$\boxed{z = -\frac{C}{2}\left(3 - 4e^{-\mathrm{i}\phi} + e^{-2\mathrm{i}\phi}\right)}. \tag{38}$$

Separating the real and imaginary parts gives

$$\begin{aligned} x &= -\frac{C}{2}\left(3 - 4\cos\phi + \cos 2\phi\right) \\ &= -C(1 - \cos\phi)^2, \end{aligned} \tag{39}$$

and

$$\boxed{y = \frac{C}{2}\left(-4\sin\phi + \sin 2\phi\right)}. \tag{40}$$

These expressions describe the location of each spatial Fourier mode in the complex z-plane as the Courant number and wave number vary.

A.2 RK2 Amplification Factor

The explicit midpoint RK2 method applied to the scalar equation

$$\frac{d\hat{T}}{dt} = \lambda \hat{T} \tag{41}$$

uses the intermediate stage

$$\hat{T}^{n+1/2} = \hat{T}^n + \frac{\Delta t}{2}\lambda \hat{T}^n, \tag{42}$$

followed by

$$\hat{T}^{n+1} = \hat{T}^n + \Delta t\, \lambda \hat{T}^{n+1/2}. \tag{43}$$

Substituting the intermediate stage into the final update gives

$$\hat{T}^{n+1} = \left(1 + \lambda\Delta t + \frac{(\lambda\Delta t)^2}{2}\right)\hat{T}^n. \tag{44}$$

Therefore, the numerical amplification factor is

$$\boxed{G_{\mathrm{RK2}}(z) = 1 + z + \frac{z^2}{2}}. \tag{45}$$

With $z = x + \mathrm{i}y$, its real and imaginary components are

$$\mathrm{Re}(G_{\mathrm{RK2}}) = 1 + x + \frac{x^2 - y^2}{2}, \tag{46}$$

$$\mathrm{Im}(G_{\mathrm{RK2}}) = y(1 + x). \tag{47}$$

The absolute-stability condition is $|G_{\mathrm{RK2}}| \leq 1$. Expressed in terms of x and y, this becomes

$$\boxed{\left(1 + x + \frac{x^2 - y^2}{2}\right)^2 + y^2(1 + x)^2 \leq 1}. \tag{48}$$

The boundary of the absolute-stability region is defined by $|G_{\mathrm{RK2}}| = 1$. Evaluating this condition over the Fourier phases and Courant numbers produces the stability diagram presented in the main text.

A.3 Limiting Courant Number

The limiting Courant number can be obtained by examining the highest-frequency mode, $\phi = \pi$. Substituting this phase into the eigenvalue expression gives

$$z = -\frac{C}{2}(3 + 4 + 1) = -4C. \tag{49}$$

The corresponding RK2 amplification factor is

$$G_{\mathrm{RK2}}(-4C) = 1 - 4C + 8C^2. \tag{50}$$

For $C \geq 0$, this expression is positive. Therefore, the stability condition reduces to

$$1 - 4C + 8C^2 \leq 1. \tag{51}$$

Rearranging gives

$$4C(2C - 1) \leq 0. \tag{52}$$

Hence, the highest-frequency mode satisfies the stability condition for

$$\boxed{0 \leq C \leq 0.5}. \tag{53}$$

Evaluation of the complete Fourier spectrum confirms that this mode determines the limiting Courant number for the second-order upwind spatial operator combined with explicit midpoint RK2. Thus, the theoretical maximum stable Courant number is $C_{\max} = 0.5$, consistent with the numerical stability study reported in Section 4.2.

B. Fourier Stability Analysis of Classical RK4 with Second-Order Upwind

B.1 Spatial Eigenvalue

The spatial discretization remains the second-order upwind scheme,

$$\frac{dT_j}{dt} = -\frac{u}{2\Delta x}\left(3T_j - 4T_{j-1} + T_{j-2}\right). \tag{54}$$

Using a Fourier mode $T_j(t) = \hat{T}(t)e^{\mathrm{i}j\phi}$ gives

$$\lambda(\phi) = -\frac{u}{2\Delta x}\left(3 - 4e^{-\mathrm{i}\phi} + e^{-2\mathrm{i}\phi}\right). \tag{55}$$

With $C = u\Delta t/\Delta x$, the dimensionless eigenvalue becomes

$$\boxed{z = \lambda\Delta t = -\frac{C}{2}\left(3 - 4e^{-\mathrm{i}\phi} + e^{-2\mathrm{i}\phi}\right)}. \tag{56}$$

Writing $z = x + \mathrm{i}y$, the real and imaginary parts are

$$\boxed{x = -\frac{C}{2}\left(3 - 4\cos\phi + \cos 2\phi\right)}. \tag{57}$$

and

$$\boxed{y = \frac{C}{2}\left(-4\sin\phi + \sin 2\phi\right)}. \tag{58}$$

B.2 RK4 Amplification Factor

The classical four-stage Runge–Kutta method has the amplification polynomial

$$\boxed{G_{\mathrm{RK4}}(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}}. \tag{59}$$

For $z = x + \mathrm{i}y$, write the amplification factor as

$$G_{\mathrm{RK4}} = A + \mathrm{i}B. \tag{60}$$

Expanding the polynomial and separating its real and imaginary parts gives

$$\boxed{\begin{aligned} A = 1 + x &+ \frac{x^2 - y^2}{2} \\ &+ \frac{x^3 - 3xy^2}{6} \\ &+ \frac{x^4 - 6x^2y^2 + y^4}{24}. \end{aligned}} \tag{61}$$

and

$$\boxed{\begin{aligned} B = y + xy &+ \frac{3x^2y - y^3}{6} \\ &+ \frac{4x^3y - 4xy^3}{24}. \end{aligned}} \tag{62}$$

The magnitude of the amplification factor is therefore

$$|G_{\mathrm{RK4}}| = \sqrt{A^2 + B^2}. \tag{63}$$

The absolute-stability condition becomes

$$\boxed{A^2 + B^2 \leq 1}. \tag{64}$$

The stability diagram is constructed by evaluating this expression over a range of Fourier phases and Courant numbers. The contour $|G_{\mathrm{RK4}}| = 1$ separates stable and unstable individual modes, while additional contours, such as $|G_{\mathrm{RK4}}| = 0.5$, indicate levels of attenuation within the stable region.

B.3 Limiting Courant Number

For the highest-frequency mode, $\phi = \pi$, the dimensionless eigenvalue reduces to

$$z = -4C. \tag{65}$$

Defining $a = 4C$, the amplification factor along the negative real axis is

$$G_{\mathrm{RK4}}(-a) = 1 - a + \frac{a^2}{2} - \frac{a^3}{6} + \frac{a^4}{24}. \tag{66}$$

The boundary of the stable interval on the negative real axis occurs at the first nonzero positive value of a for which $G_{\mathrm{RK4}}(-a) = 1$. Thus,

$$-a + \frac{a^2}{2} - \frac{a^3}{6} + \frac{a^4}{24} = 0. \tag{67}$$

Multiplying by 24 and factoring out a gives

$$a\left(a^3 - 4a^2 + 12a - 24\right) = 0. \tag{68}$$

The first nonzero positive root is approximately

$$a_{\max} \approx 2.7853. \tag{69}$$

Consequently,

$$C_{\max} = \frac{a_{\max}}{4} \approx 0.6963. \tag{70}$$

The complete Fourier-mode analysis confirms that the highest-frequency mode determines the limiting Courant number for the second-order upwind spatial operator combined with classical RK4. Hence,

$$\boxed{C_{\max} \approx 0.6963}. \tag{71}$$

This theoretical limit agrees with the sampled stability diagram, subject to the finite resolution of the numerical parameter grid.

C. Preliminary ODE Verification of RK2

C.1 Governing Equation and Analytical Solution

Before solving the advection equation, the time-integration method was tested using the non-homogeneous ordinary differential equation

$$\frac{dv}{dt} = -v + \sin t, \qquad v(0) = 1. \tag{72}$$

The equation can be written as

$$\frac{dv}{dt} + v = \sin t. \tag{73}$$

Solving this first-order linear ODE gives the analytical solution

$$\boxed{v(t) = \frac{\sin t - \cos t}{2} + \frac{3}{2}e^{-t}}. \tag{74}$$

This solution satisfies the initial condition $v(0) = 1$ and provides an exact reference for evaluating the temporal discretization independently of spatial error.

C.2 Explicit Midpoint RK2 Formulation

Applying the explicit midpoint RK2 method to the ODE gives the intermediate estimate

$$v^{n+1/2} = v^n + \frac{\Delta t}{2}\left(-v^n + \sin t_n\right), \tag{75}$$

followed by the full time-step update

$$v^{n+1} = v^n + \Delta t\left[-v^{n+1/2} + \sin\left(t_n + \frac{\Delta t}{2}\right)\right]. \tag{76}$$

Starting from $v^0 = 1$, these equations are applied successively until the prescribed final time is reached. At each step, the numerical solution is compared with the analytical solution evaluated at the same time.

The error measure used in the original implementation is the mean absolute error over the time points,

$$E_{\mathrm{MAE}} = \frac{1}{N+1}\sum_{n=0}^{N}\left|v^n - v(t_n)\right|, \tag{77}$$

where N is the number of time steps. This is a normalized discrete L^1-type error measure rather than an unnormalized integral norm.

C.3 Temporal Stability

To examine stability independently of the forcing term, consider the homogeneous equation

$$\frac{dv}{dt} = -v. \tag{78}$$

Applying the two RK2 stages yields

$$v^{n+1} = \left(1 - \Delta t + \frac{\Delta t^2}{2}\right) v^n. \tag{79}$$

The amplification factor is therefore

$$G = 1 - \Delta t + \frac{\Delta t^2}{2}. \tag{80}$$

For positive Δt, this expression is positive. The stability condition $|G| \leq 1$ consequently reduces to

$$1 - \Delta t + \frac{\Delta t^2}{2} \leq 1, \tag{81}$$

which gives

$$\boxed{0 < \Delta t \leq 2}. \tag{82}$$

Thus, the homogeneous ODE remains absolutely stable for time steps up to $\Delta t = 2$, with the upper limit corresponding to marginal stability.

C.4 Temporal Convergence

The temporal convergence study was performed using successively refined time steps,

$$\Delta t \in \left\{ \frac{1}{5}, \frac{1}{10}, \frac{1}{20}, \frac{1}{40}, \frac{1}{80} \right\}. \tag{83}$$

At each resolution, the numerical and analytical solutions were evaluated at the same time points, and the mean absolute error was calculated. The errors were plotted against the number of time steps on logarithmic axes.

For a method of temporal order p, the error is expected to scale as

$$E \propto (\Delta t)^p. \tag{84}$$

Since the number of time steps is proportional to $1/\Delta t$ for a fixed final time, this relationship can also be written as

$$E \propto N^{-p}. \tag{85}$$

A log–log plot of error against the number of time steps should therefore have a slope of approximately $-p$. The observed slope was consistent with second-order temporal convergence, supporting the expected accuracy of the explicit midpoint RK2 method. The original study also reported a mean absolute error of approximately 0.002 for $\Delta t = 1/5$.

This preliminary verification provides a reference for the subsequent PDE simulations, where temporal discretization is combined with spatial discretization and additional errors arise from the finite-volume approximation.